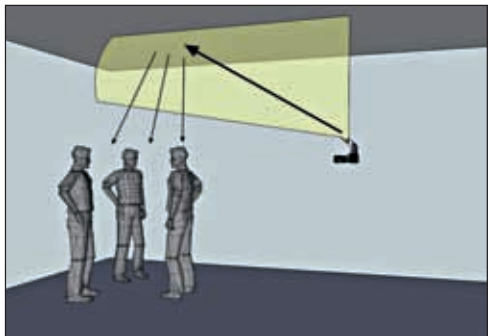


situations where ambient light may be enough but a desired effect from the flash is needed. Aperture and Shutter Priority modes treat it as a fill flash (concept explained further ahead). And finally in Manual mode since all settings are under your control, the camera exposes ambient light based on how you set the aperture, shutter speed and ISO, allowing you to have any desired flash ratio.

Flash Exposure Compensation is another concept you'll encounter commonly, which is the key to changing the flash ratio. Exposure Compensation basically tells the camera to take whatever flash intensity it was going to use, and to override that by the FEC setting. The big difference is that while EC affects the exposures for both flash and ambient light, FEC only affects flash intensity. Most current SLR flash systems employ some form of through-the-lens (TTL) metering. Digital TTL flash metering works by bouncing one or more tiny pre-flash pulses off the subject immediately before the exposure begins, which are then used to estimate what flash intensity is needed during the actual exposure.

Now we'll see some important techniques in clicking with flashes. The first concept to understand is light distribution, which is how much contrast the image will have, or how much will the difference be between the lightest and darkest portions of the subject. This also describes what photographers call 'soft light', which is more scattered or originating from a larger area, and 'hard light', which is more directional and concentrated. When using a localized light source, the side facing the source will be intensely lit while the opposing side will be nearly dark as it receives no direct light but only that which is bounced off surrounding objects.

In comparison, in a distributed light source the highlights and shadows seem to appear softer because light is hitting the object from a wider angle in this situation. Conceptually grasping this concept is not tough, but realizing the practical implications takes more understanding. Less contrast can make pictures of



Flash Bounce Visualization.